

PROBLEM STATEMENT: VECTOR ADDITION

```
%%writefile vector_add.cu

#include <iostream>

#include <cuda_runtime.h>

using namespace std;

__global__ void vectorAdd(int *A, int *B, int *C, int n) {

    int i = blockIdx.x * blockDim.x + threadIdx.x;

    if (i < n) {

        C[i] = A[i] + B[i];

    }

}

int main() {

    int n;

    cout << "Enter size of vector: ";

    cin >> n;

    int size = n * sizeof(int);

    int *h_A = new int[n];

    int *h_B = new int[n];

    int *h_C = new int[n];

    cout << "Enter elements of Vector A:\n";

    for(int i = 0; i < n; i++) {
```

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        cin >> h_A[i];
    }

    cout << "Enter elements of Vector B:\n";
    for(int i = 0; i < n; i++) {
        cin >> h_B[i];
    }

    int *d_A, *d_B, *d_C;

    cudaMalloc((void**)&d_A, size);
    cudaMalloc((void**)&d_B, size);
    cudaMalloc((void**)&d_C, size);

    cudaMemcpy(d_A, h_A, size, cudaMemcpyHostToDevice);
    cudaMemcpy(d_B, h_B, size, cudaMemcpyHostToDevice);

    int threadsPerBlock = 256;
    int blocksPerGrid = (n + threadsPerBlock - 1) / threadsPerBlock;

    vectorAdd<<<blocksPerGrid, threadsPerBlock>>>(d_A, d_B, d_C, n);

    cudaMemcpy(h_C, d_C, size, cudaMemcpyDeviceToHost);

    cout << "\nResult Vector:\n";
    for(int i = 0; i < n; i++) {
        cout << h_C[i] << " ";
    }

```

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    cudaFree(d_A);

    cudaFree(d_B);

    cudaFree(d_C);


    delete[] h_A;

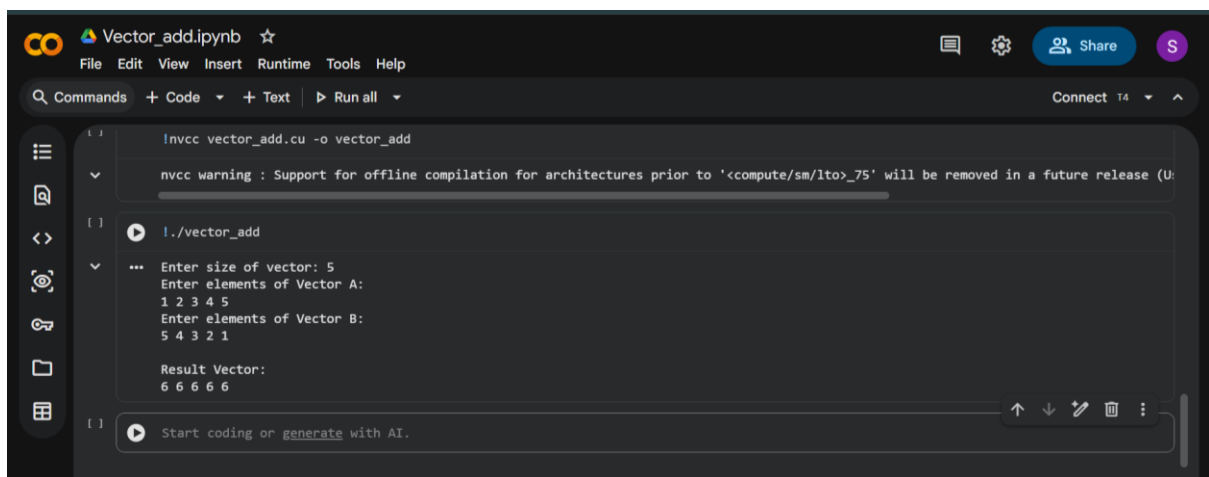
    delete[] h_B;

    delete[] h_C;


    return 0;
}

```

OUTPUT:



```

Vector_add.ipynb ☆
File Edit View Insert Runtime Tools Help
Commands + Code + Text Run all
Connect T4 ^

[] In[ ]: !nvcc vector_add.cu -o vector_add
nvcc warning : Support for offline compilation for architectures prior to 'compute/sm/lto>_75' will be removed in a future release (U

[] In[ ]: !./vector_add
...
Enter size of vector: 5
Enter elements of Vector A:
1 2 3 4 5
Enter elements of Vector B:
5 4 3 2 1
Result Vector:
6 6 6 6 6

[] In[ ]: Start coding or generate with AI.

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%%writefile parallel_gpu.cu
```

```
# Compile the CUDA code
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```
!nvcc parallel_gpu.cu -o parallel_gpu
```

```
# Run the program
```

```
!./parallel_gpu
```